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MILITARY SPECIFICATION

INTERCOMMUNICATION STATION AND LOUDSPEAKER, MAGNETIC (INTERCOMMUNICATING STATION LS-130( )/FI AND LOUDSPEAKER LS-114( )/U)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers two types of equipment designated as Intercommunicating Station LS-130( )/FI and Loudspeaker LS-114( )/U.

1.1.1 Intercommunicating Station LS-130( )/FI is a remote (speaker-microphone) station with a call switch housed in a wooden enclosure.

1.1.2 Loudspeaker LS-114( )/U is a permanent magnet type loudspeaker with an impedance matching transformer housed in a wooden enclosure.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

C-F-206 — Felt, Mechanical, Roll.

NN-B-621 — Boxes; Wood, Nailed and Lock Corner.

QQ-S-571 — Solder; Soft (Tin, Tin-Lead, and Lead-Silver).

TT-L-57 — Lacquer, Rubbing, Clear (For Wood Furniture).

TT-S-190 — Sealer, Sanding, Lacquer Type (For Wood Furniture).

TT-S-711 — Stain; Oil Type, Wood, Interior.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-M-14 — Molding Plastics and Molded Plastic Parts, Thermosetting.

MIL-T-27 — Transformers and Inductors (Audio, Power, Pulse)

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MIL–P–66 — Plywood; Flat Panel.
MIL–W–76 — Wire and Cable, Hook-up, Electrical Insulated.
JAN–P–106 — Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
JAN–P–108 — Packaging and Packing For Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior.
MIL–P–116 — Preservation, Methods of.
JAN–P–127 — Packaging and Packing For Overseas Shipment — Tape, Adhesive, Pressure Sensitive, Water Water Resistant.
MIL–T–152 — Treatment, Moisture and Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL–A–397 — Adhesive, Room-Temperature and Intermediate-Temperature Setting Resin.
MIL–M–2312 — Mildew - Resistance and Moisture-Resistance Treatment For Felt, Wool.
MIL–P–3115 — Plastic-Material, Laminated, Thermosetting, Sheets, Paper-Base, Phenolic Resin.
MIL–A–5433 — Adhesive; Application of Room Temperature and Intermediate Temperature Setting Resin.
MIL–M–13231 — Marking of Electronic Items.
MIL–F–14072 — Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 — Inspection Terms And Definitions.
MIL–STD–129 — Marking for Shipment And Storage.
MIL–STD–169 — Extreme Temperature Cycle.
MIL–STD–170 — Moisture - Resistance Test Cycle For Ground Signal Equipment.
MIL–STD–252 — Wired Equipment, Classification of Visual and Mechanical Defects For.

DRAWINGS

GNAL CORPS

56–1 — Plywood.
56–2 — Wood, Low Density.
SC–D–16286 — Standard Cycle For Moisture Resistance Tests of Component Parts.
SC–DL–98123 — Intercommunicating Station LS–130( )/FI.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

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2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washington 6, D. C.

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification.

(Application for copies should be addressed to the Association of American Railroads, 202 Chicago Union Station, Chicago 6, Ill.)

NATIONAL BUREAU OF STANDARDS

Handbook H–28 — Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Procurement model.** The procurement model of Loudspeaker LS–114( )/U shall be the one which will be available for inspection by prospective bidders and will be lent to the contractor, in accordance with the invitation for bids. Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

3.1.1 *Construction.* Physical construction of Loudspeaker LS–114( )/U shall conform to the procurement model.

3.1.2 *Features.* Loudspeaker LS–114( )/U shall incorporate all features of the procurement model.

3.1.3 *Verification.* As soon as the procurement model of Loudspeaker LS–114( )/U is received, and with the minimum necessary adjustment and alinement (if any), the contractor shall measure its performance for compliance with specified characteristics of the equipment on contract. If the model does not meet the specified performance characteristics, the contractor shall adjust or aline it for optimum performance and then repeat the measurements. The contractor shall immediately prepare copies of a test report on the measurements, and forward them as required by the contract. In case of conflict between specified performance characteristics and performance of the model, the former shall govern.

3.2 **Construction.** Intercommunicating Station LS–130( )/FI shall be constructed in accordance with the drawings on Drawing and Data List SC–DL–98123 and shall meet the requirements of this specification. In case of conflict between the drawings and specification, the specification shall govern.

3.3 **Preproduction samples.**

3.3.1 *Quantities.* Preproduction samples shall be furnished for Government inspection and approval, as listed below:

(a) Two complete equipments

(b) Two switches (See 3.18.2.2)

(c) Two loudspeaker units

3.3.2 *Fabrication and assembly.* Preproduction samples shall be fabricated by the contractor and shall meet all requirements specified for the equipment on contract. The preproduction samples shall be assembled using parts, materials, and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

3.3.3. *Approval and use.* Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.3.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

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3.3.4 *Reference standards.* After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.4 Plastic material. Where not machined, plastic material shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the kind of material used and the intended application. Plastic material shall conform to Specification MIL–M–14 and MIL–P–3115.

3.5 Wood. Solid wood shall conform to Drawing 56–2.

3.6 Plywood. Plywood shall be 5 ply hardwood plywood conforming to Specification MIL–P–66 and Drawing 56–1. Plywood shall be regular sanded both sides.

3.7 Glues and gluing. Wood-to-wood joints shall be made with glue conforming to Specification MIL–A–397, and the glue shall be applied in accordance with Specification MIL–A–5433

3.8 Soldering.

3.8.1 *Solder.* Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ–S–571.

3.8.2 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.8.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.9 Cleaning. Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.10 Wire, hookup. Wherever practicable, hookup wire shall be stranded. Conductor size shall be sufficient to avoid deterioration in electrical properties of the wire from overheating. All hookup wire shall be type MW in accordance with Specification MIL–W–76.

3.11 Wiring. All wiring shall be neat and sturdy and as short as possible but sufficient slack shall be provided for field repair of broken connections.

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3.12 **Felt.** Felt pads shall conform to Specification C–F–206 and shall be given Type III treatment in accordance with Specification MIL–M–2312.

3.13 **Finishes.** Finishes shall be in accordance with Specification MIL–F–14072.

3.14 **Color.** The cases shall have a dark walnut finish.

3.14.1 After finish sanding all exposed surfaces shall be treated with a walnut colored stain filler in accordance with Specification TT–S–711.

3.14.2 One coat of clear sealer in accordance with Specification TT–S–190 shall be applied after the filler has set and been smoothed.

3.14.3 Two coats of clear lacquer in accordance with Specification TT–L–57 shall be applied over the sealed surfaces after allowing the sealer to set and be smoothed.

3.14.4 All exposed surfaces shall be rubbed to produce a smooth surface.

3.14.5 All non-exposed surfaces shall be smooth sanded and coated with one coat of clear sealer in accordance with Specification TT–S–190.

3.14.6 After the sealer has set, all non-exposed surfaces shall have one coat of clear lacquer in accordance with TT–L–57.

3.15 **Tropicalization.**

3.15.1 *Treatment of assembled equipment.* Assembled equipment shall be treated in accordance with Specification MIL–T–152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc.) which is applicable on contract.

(2) It does not incorporate any material listed in 3.15.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL–F–14072.

3.15.2 *Treatment of materials.*

3.15.2.1 *Treating materials.* Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.15.2.2. *Toxicity.* Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.15.2.3 *Flexibility.* Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.15.2.4 *Statement of treatment.* The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL–T–152, the above statement shall be incorporated in

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the statement of treatment required by that specification.

3.15.3 *Materials not resistant to moisture and fungi.* The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.
(d) Felt, hair or wool.
(e) Jute.
(f) Leather.
(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.16 Securing of parts.

3.16.1 *General.* Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain components and parts securely when the equipment is subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the equipment is subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of equipment performance below full specification requirements. Where practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.17 Screw threads.

3.17.1 *Form and fit.* Unless otherwise specified, screw threads shall conform to Handbook H28 and supplement thereto; the Unified Thread Form and Series shall be class 2A or 2B fit, and the American National Thread Series shall be class 2 fit.

3.18 Electrical requirements.

3.18.1 *Common requirements for the loudspeaker units of Intercommunicating Station LS–130( )/FI and Loudspeaker LS–114 ( )/U.*

3.18.1.1 *Frequency response.* At full rated output over the range of 250 to 5000 c.p.s. the average response shall be not less than 95 db relative to .0002 dynes/cm^2. The response shall be flat within 14 db over the above frequency range.

3.18.1.2 *Distortion and power handling capacity.* With 2 watts electrical input the harmonic distortion shall be not greater than 5% at frequencies of 1000 to 5000 c.p.s. and for frequencies of 400 to 1000 c.p.s. the distortion shall be not greater than 10%. The loudspeaker shall have an electrical power handling capacity of 2 watts.

3.18.1.3 *Impedance.* The impedance of the voice coils shall be 45 ohms ±15% at 1000 c.p.s.

3.18.1.4 *Dielectric strength.* The loudspeaker units shall withstand a peak voltage of 100 volts 60 cycles a.c. applied between voice coil and frame for a period of one minute with a simultaneous excitation of the voice coil by a two watt 60 c.p.s. signal.

3.18.2 *Intercommunicating Station LS–130 ( )/FI.*

3.18.2.1 *Microphone output.* The loudspeaker when used as a microphone and connected to a 50 ohm non-inductive load resistor and tested in a sound field of 10 dynes per sq. cm. shall develop the following minimum voltages:

Frequency	*Minimum Voltage*
250 to 500 c.p.s.	5.5 Millivolts
500 to 1000 c.p.s.	2.5 Millivolts
1000 to 5000 c.p.s.	0.75 Millivolts

3.18.2.2 *Calling switch.* The calling switch shall be Switchcraft Type No. 1000, or equal with soldered terminals, or equal, and shall be capable of passing the life test of 4.11.7.

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3.18.2.2.1 The switch shall be wired to one pair of terminals and the loudspeaker shall be wired to a second pair of terminals on a four pair terminal block Type No. 4–10 as manufactured by Howard B. Jones, Chicago, Ill., or equal. Each pair shall be properly identified.

3.18.3 *Loudspeaker LS–114( )/U.*

3.18.3.1 *Matching transformer.* An impedance matching transformer, in accordance with the requirements of Specification MIL–T–27 for Grade 1, Class A transformers, shall be provided. The transformer shall have a power handling capacity of 2 watts over the frequency range of 250 to 5000 c.p.s. The secondary winding of the transformer shall be so tapped as to present impedances of 83,000; 20,830; and 5,210 ohms, respectively, to the primary terminals, when a 45 ohm resistor is connected, in turn, across each secondary tap. Transformer should be mounted so its circuit marking is clearly visible when the back of case is removed.

3.18.3.1.1 *Circuit wiring.* The primary and secondary windings of the transformer shall be wired to one side of a six pair terminal block, as manufactured by H. B. Jones Type No. 6-10, or equal, with leads color coded in accordance with the model. The loudspeaker unit, unless otherwise specified in the bid request or contract shall be wired at the terminal block to the secondary tap which will provide a primary impedance of 20,830 ohms.

3.19 **Size.** The overall dimensions of the equipments covered by this specification shall not exceed 7½ inches in height by 4½ inches in depth by 7⅜ inches long.

3.20 **Service conditions.** The equipment shall meet the following service conditions:

3.20.1 *Equipment operating.* The equipment shall meet the requirements of this specification under any probable combination of the following:

(a) *Temperature.* Any ambient temperature in the range of –65°F. to 150°F., including sun load.

(b) *Elevation.* Any elevation up to 10,000 feet above sea level.

(c) *Relative humidity.* Any relative humidity up to 95 percent.

(d) *Orientation.* Any orientation up to 20 degrees from the normal operating position.

3.20.2 *Equipment non-operating.* The equipment shall meet the requirements of 3.20.1 after subjection to any one or more of the following non-operating conditions:

(a) *Temperature.* Continuous exposure for 72 hours at 160°F. and 72 hours at –80°F. at any orientation.

(b) *Elevation.* Elevations up to 50,000 feet above sea level.

(c) *Relative humidity.* Relative humidity up to 100 percent including condensation caused by temperature changes.

(d) *Orientation.* Indefinite storage in any orientation.

(e) *Vibration, bounce and shock.* (Sec 3.20.2.1)

3.20.2.1 *Vibration, bounce and shock.*

3.20.2.1.1 *Vibration.* Except for internal resonance of parts and subassemblies specified herein, the equipment shall have no mechanical resonance below 55 cycles per second.

3.20.2.1.2 *Bounce and shock.* The equipment shall meet full specification requirements upon completion of the tests of 4.11.3 and 4.11.4.

3.21 **Interchangeability.** Corresponding components, replaceable subassemblies, and replaceable parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance

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with the foregoing. If the contractor is in doubt as to whether a particular subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.22 **Marking.**

3.22.1 *General.* Marking shall conform to Specification MIL–M–13231.

3.22.2 *Visibility.* Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.23 **Workmanship.** The equipments shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein:

3.5 Wood.
3.6 Plywood.
3.7 Glues and gluing.
3.8 Flux and cleaning agents for soldering.
3.8.3. Process of soldering.
3.9 Cleaning of equipment.
3.11 Wiring.
3.16.1 General requirements for securing of parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Inspection, general.** When MIL–STD–105 specifies an action by the Government it shall at the option of the Government be performed either by the Government or by the contractor under the supervision of the Government inspector with final responsibility for acceptance of material resting with the Government inspector.

4.1.1 *Definitions.* Standard MIL–STD–109 applies for definitions of inspection terms used herein.

4.1.2 *Classification of inspection.* Inspection shall be classified as follows:

(a) Preproduction inspection

(b) Acceptance inspection

4.2 **Preproduction inspection.** Preproduction inspection shall consist of the group A inspection and nondestructive group B inspection specified in tables II, and III and the preproduction inspection specified in table I. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, and (3) shock, drop; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE I. *Preproduction inspection*

Inspection (For additional preproduction inspection see 4.2)	Reqt Par.	Insp Par.
Temperature	3.20	4.11.1
Vibration	3.20.1.1	4.11.2
Bounce	3.20.2.1	4.11.3
Shock	3.20.2.1	4.11.4
Moisture resistance	3.20	4.11.5 and 4.11.6
Switch life test	3.18.2.2	4.11.7

4.3 **Acceptance inspection.** Acceptance inspection shall include group A and group B inspection as specified in 4.3.1 through 4.3.2.2. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B inspection. Each unit which will be subjected to group A and group B inspection shall be preconditioned after final assembly. (See 4.4)

4.3.1 *Group A inspection.* Group A inspection (including sampling) shall conform to table II and Standard MIL–STD–105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

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TABLE II. *Government group A inspection*

Inspection	Rqt. Par.	Insp. Par.	AQL		
			Major	Minor	Control
Intercommunicating Station LS–130( )/FI					
Visual and mechanical	3.23	4.5	1.5 dphu	6.5 dphu	25 dphu*
Dielectric	3.18.1.4	4.6	1.5 dphu for group		
Loudspeaker response	3.18.1.1	4.7.1			
Microphone response	3.18.2.1	4.7.2			
Impedance	3.18.1.3	4.8			
Loudspeaker LS–114( )/U					
Visual and mechanical	3.23	4.5	1%	4%	25 dphu
Dielectric	3.18.1.4	4.6	1% for group		
Loudspeaker response	3.18.1.1	4.7.1			
Impedance	3.18.1.3	4.8			

* Dphu—Defects per hundred units.

4.3.2 *Group B inspection.* Group B inspection (including sampling) shall conform to table III and the Appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. Disposition of rejected product (sample units and inspection lots) shall be in accordance wih Standard MIL–STD–105 and 4.3.4.

4.3.2.1 *Group B sampling plans.* The group B sampling plans, as listed in table III, shall be as follows:

Group B	AQL	Inspection level for normal inspection	Inspection level for reduced inspection
B–2	6.5%	L–7	L–5

4.3.2.2 *Procedure in case of failure.* When an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

TABLE III. *Government group B inspection*

Inspection	Rqt. Par.	Insp. Par.	Group B Plans (See 4.3.2.1)
Intercommunicating Station LS–130( )/FI			
Distortion	3.18.1.2	4.9	B–2
Dimensions (Interchangeability)	3.21	4.10	B–2
Loudspeaker LS–114( )/U			
Distortion	3.18.1.2	4.9	B–2
Dimensions (Interchangeability)	3.21	4.10	B–2

4.3.3 *Reinspection of conforming group B sample units.* Unless otherwise specified, sample units which have been subjected to and passed group B inspection, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.4 *Disposition of nonconforming product.* When defective sample units or rejected

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inspection lots are resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault.

4.4 Bounce preconditioning. The equipment shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the equipment shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.5 Visual and mechanical inspection. Parts and equipment shall be examined for the defects listed in Standard MIL–STD–252.

4.6 Dielectric test. The loudspeaker units shall be tested for conformance with 3.18.1.4.

4.7 Frequency response tests. Loudspeaker unit mounted in their case shall be tested in substantially free sound field. Tests made in the acoustic test chamber of the Signal Corps Engineering Laboratories shall be considered as standard.

4.7.1 *Loudspeaker response.* Response measurements of loudspeaker units shall be made at a distance of three feet on the axis of the loudspeaker. A constant voltage of 9.5 volts shall be impressed on the speaker voice coil and the frequency varied over the range of 250 to 5000 c.p.s. The response shall meet the requirements of 3.18.1.1.

4.7.2 *Microphone response (LS–130( )/FI only).* The loudspeaker shall be tested as microphones to show compliance with 3.18.2.1.

4.8 Impedance test. The loudspeaker unit shall be connected in series with a variable resistance across a low impedance signal source of 1000 c.p.s. The voltages drop across the loudspeaker voice coil and across the variable resistance shall be measured with a vacuum tube voltmeter, Ballantine Model 300, or equal. The value of the resistance shall be varied until the voltage drops measured across the voice coil and resistance are equal. The value of the speaker voice coil shall be then considered to be numerically equal to the value of the resistance. The impedance shall be in compliance with 3.18.1.3.

4.9 Distortion and power handling capacity. The loudspeaker units shall be tested for compliance with 3.18.1.2.

4.10 Inspection for interchangeability. The dimensions listed below, as applicable, shall be gaged or measured (see table III) to determine compliance with the physical interchangeability requirement of 3.21. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies when such dimensions affect acceptability of of insertable subassemblies.

(c) Location of fasteners on separable parts or assemblies which must mate together, such as cases and covers.

(d) Location of fasteners which receive mating parts of major components.

4.11 Service condition tests.

4.11.1 *Temperature.* The equipment shall be subjected to the temperature cycle shown on Standard MIL–STD–169. It shall meet

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full specification requirements when measured at steps 3, 10 and 7.

4.11.2 *Vibration test.* The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.20.2.1.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at $\frac{1}{64}$ inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.11.3 *Bounce test; light equipment.* The equipment shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 r.p.m. ± 1 percent, for a total of 3 hours. During each ½ hour period of the test, the equipment shall rest on a different face. After test the equipment shall meet full specification requirements and there shall be no damage except surface abrasion.

4.11.4 *Shock test drop.* The equipment shall be dropped a distance of 1 foot on each corner, edge, and face (total of 26 drops). Drops shall be made from a quick-release hook; and Acme Drop Tester, or equal; a Conbur Tester (10-degree inclined plane), or equal; or a basket-type pendulum tester. The floor or barrier receiving the impact shall be 2-inch fir backed by concrete or a rigid steel frame. After test the equipment shall be operable. Minor mechanical damage is permissible.

4.11.5 *Moisture-resistance test for equipment.* The equipment shall be tested as follows:

4.11.5.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.11.5.2 *Test method.*

(a) Dry at 130° ± 5° F. for 24 hours.

(b) Condition at 77° ± 5° F. and 40 to 50 percent relative humidity for 24 hours.

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(c) Take measurements as specified in 4.7.1 and 4.7.2 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL–STD–170. The measurements specified in 4.7.1 and 4.7.2 shall be made at the conclusion of the 5th cycle and shall comply with 4.7.1 and 4.7.2.

4.11.6 *Moisture resistance test for part.* The sample (3.3.1) loudspeaker units shall be subjected to a 10 cycle humidity test. Each cycle shall be in accordance with Drawing SC–D–16286 (Vibration Test, Note 2 of Drawing, shall not apply). A listening test for both speech and music supplied from a good source shall be made on the loudspeaker at the rated output to determine the quality of the loudspeaker at the end of each cycle. The units shall not be removed from the test chamber for testing.

4.11.7 *Switch life test.* The sample (3.3.1) calling switches (push button) will be subjected to an accelerated life test. The plunger will be operated 100,000 times, at a rate of 10 operations per minute, with contacts connected in an energized circuit equivalent to that used in the equipment. One operation shall consist of depressing the plunger down to a distance of its normal travel and allowance for the plunger to return to its original position. The switch shall not fail electrically or mechanically during the test. After 100,000 successful operations, the switch will be inspected. Any sticking of contacts or visible damage to part of the switch shall be considered failure to pass the test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Intercommunicating Station LS–130( )/FI and Loudspeaker LS–114 ( )/U, shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and further described herein.

5.1.1.1 *Intercommunicating Station LS–130( )FI and Loudspeaker LS–114( )/U.* Each intercommunication station and each loudspeaker shall be packaged individually Method IC–5 as follows: Cushion each item on all surfaces with cells or pads or both fabricated of double face corrugated fiberboard having a minimum dry bursting strength of 175 pounds, and designed to absorb the shock of impact encountered in handling and transit. Place the cushioned item within a close-fitting regular slotted style corrugated fiberboard box. Grade 5 conforming to Specification JAN–P–108. Box closure shall be in conformance with the referenced box specification further reinforced at all closures, corners and joints by means of Type I, Grade A, olive drab tape, conforming to Specification JAN–P–127. Packages prepared as above, shall be required to pass the rough handling and the water-spray test prescribed in Specification MIL–P–116.

5.1.2 *Level C.* Intercommunicating Station LS–130( )/FI and Loudspeaker LS–114 ( )/U shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration, and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* A suitable quantity of packed items bearing the same stock number shall be packed together in shipping containers conforming to specification JAN–P–106 or PPP–B–601. The gross weight of any one shipping container shall not exceed 150 pounds. Box closure shall be in conformance with the appendix to the box specification.

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5.2.1.1 *Metal strapping.* Metal strapping shall be applied in accordance with the requirements of the appendix of the box specification, only for direct shipment to ports.

5.2.2 *Level B.* A suitable quantity of packed items bearing the same stock number shall be packed together in shipping containers conforming to PPP–B–621, LLL–B–631, LLL–B–636 and PPP–B–601.

5.2.3 *Level C.* Intercommunicating Station LS–130( )/FI and Loudspeaker LS–114 ( )/U, shall be packed for shipment in a manner conforming to the requirements of Uniform Freight classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Military Standard, MIL–STD–129.

6. NOTES

6.1 **Intended use.** Intercommunicating Station LS–130( )/FI is intended for use in intercommunication station by operating the call switch and using the loudspeaker of the station unit as a microphone.

6.1.1 Loudspeaker LS–114( )/U is intended for use as a paging loudspeaker.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Levels of packaging and packing required for shipment. (Level A, Level B, or Level C.)

(d) Samples required. (See 3.3.1).

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) The statement of treatment referenced in 3.15.2.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 **Nomenclature.** The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example LS–114W/U. As soon as possible after the award of the contract, the contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1.)

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